JWALITH MADHYAHNAPU

jwalith4418@gmail.com | +91 93905 12975 | github.com/jwalith4418-Mj | linkedin.com/in/madhyahnapu-jwalith

SUMMARY

AI/ML and software engineering student pursuing dual degrees at ABV-IIITM Gwalior and IIT Madras. Building agricultural AI, decision intelligence, formal verification, and full-stack research tools with reproducible evaluation. Google Student Ambassador; AI Synergy Hackathon 2026 winner.

EDUCATION

ABV - Indian Institute of Information Technology & Management, Gwalior **Expected 2030**
Integrated B.Tech. + M.Tech., Information Technology | JEE Main 2025: **99.16 percentile (Top 1%)**

Indian Institute of Technology Madras **Ongoing**
Bachelor of Science (BS), Aeronautical & Space Technology

EXPERIENCE

Google Student Ambassador | Google **2026 - Present**
- Selected for the 2026 program to engage students on Google technologies and AI initiatives.
- Team shortlisted for Google Cloud **Build with AI: Code for Communities** Virtual Finale; presented the solution to judges.

Robotics & AI Industrial Trainee | My Equation **Nov 2025 - Dec 2025**
- Built Python/ROS simulations across **3 mobility scenarios** with sensor-fusion pipelines to test autonomous control behavior.
- Completed 45-day industrial training with major and minor projects; received an endorsement in Robotics and AI.

PROJECTS

Blackbox - Sequential Reciprocal Matching & Research Platform **Prototype**
Python, TypeScript, JavaScript, Node.js, SQLite, libSQL, Automated Testing
- Built an interactive **200-member** simulator with budgeted clarification, reciprocal constraint checks, greedy matching, and pair-level explanations; distinguished missing information from incompatibility.
- Implemented session APIs with ownership isolation, atomic revision checks, and idempotent requests; kept hidden simulator state and future feedback server-side.
- Compared conservative and coordinated policies over **240 paired simulated worlds (480 runs)**; retained baselines, ablations, waiting-time analysis, and reproducible experiment artifacts.

CropDoctor - Agricultural AI & Decision Support **Live Demo**
Python, FastAPI, EfficientNet-Lite0, ONNX Runtime, Twilio, Azure Neural Speech, Sarvam AI
- Developed a crop-health prototype combining EfficientNet-Lite0/ONNX inference, multilingual voice workflows, and **physics-informed irrigation reasoning**.
- Implemented model lineage, **Page-Hinkley drift detection**, Brier-score calibration tracking, and a human-in-the-loop labeling queue; checked root-zone water balance under synthetic dry, rainy, and saturation scenarios.

NEXUS - AI Decision Twin & Reality Intelligence **Live Demo**
Python, FastAPI, JavaScript, RAG, Gemini API, ChromaDB, Vercel
- Built a decision-intelligence prototype connecting future exploration, strategic assumptions, decision graphs, and **tamper-evident prediction records**.
- Implemented retrieval-grounded agent workflows and prediction verification; a focused ledger check detected an altered prediction record.

VERITAS - Engineering Requirement Verification **Prototype**
Python, FastAPI, Exact Rational Arithmetic, Constraint Solving, CLI, Checkable Certificates
- Built deterministic constraint checks with guarded AI proposals and **independently checkable certificates**; exposed verification through CLI and API interfaces.
- Checked contradictory and satisfiable rational bounds; revalidated contradiction cores and satisfying witnesses using the core checker.

Additional projects: CDIS (explainable career rules and audit/replay); Aletheia (offline retrieval and selective answering); AttendanceTrackerPro (Android/Firebase); MEDHA (LLM prompt chaining, in development).

ACHIEVEMENTS

- **Winner**, AI Synergy Hackathon - Gwalior AI Summit 2026; **Virtual Finale Shortlist**, Google Cloud Build with AI, 2026.
- **Round 2 selection**, The Sequential Matching Problem, 2026; **JEE Main 2025: 99.16 percentile**.
- Participant, Bharatiya Antariksh Hackathon 2026 (ISRO x Hack2Skill) and Adobe University Hackathon 2026.

SELECTED PROGRAMS

- **McKinsey.org Forward - Selected Participant, 2026.**
- **GCI World 2026 - Selected Participant, September 2026.**

CERTIFICATIONS

- **Oracle Cloud Infrastructure 2025 Certified AI Foundations Associate** - Oracle, Jul 2025
- **AI Fluency: Framework & Foundations** - Anthropic, 2026
- Oracle Fusion Cloud Applications HCM Process Essentials Certified - Rel 1 - Oracle, Jun 2025
- AI Readiness Certificate - Gwalior AI Summit 2026 (Human-AI Delta Score: 8.06, "The Orchestrator")

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, C++, C, Java, SQL
AI/ML: Transfer Learning, ONNX, Model Calibration, Drift Detection, RAG, ChromaDB, Multi-Agent Orchestration
Backend & Data: FastAPI, REST APIs, Node.js, SQLite, libSQL, Authentication, Rule Engines, Audit Logging
Engineering & Tools: Constraint Verification, Automated Testing, Git, Vercel, OCI, ROS, Sensor Fusion, Simulation